A small JEPA world model on a low-cost arm: the encoder reads the outcome, the predictor adds nothing to it

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Abstract. Learned world models are increasingly proposed as general-purpose components of a robot learning stack: a predictor that imagines future latents can be used to rank candidate action chunks, to anticipate the outcome of a grasp, to plan over several time scales and to flag failures when the observed latent drifts from the predicted one. Whether these uses hold on a small model trained on a low-cost arm has, to the best of our knowledge, not been examined. We train a JEPA world model in the LeWorldModel recipe for $3.30 of rented GPU time on teleoperated demonstrations of one tabletop task on an SO-101 arm, and we re-test on it every published use we could find, each against a threshold committed to a public register before the result existed. Our findings separate the two halves of the model. On policy rollouts the model never saw, a linear probe on the encoder's last-frame latent separates success from failure about as well as a hand-built colour judge, and substantially better than a frozen DINOv2 encoder or the commanded joints. The predictor, however, adds nothing to this: at the grasp, the imagined latent reads the outcome at 0.70 where the current latent reads it at 0.82, and every use that requires rolling the predictor forward misses its threshold. Retraining the same recipe on 45 to 2,909 episodes, which adds a second dataset and then another lab's rollouts, does not change this picture. The controls are exploratory and come from one rig and one model per size.

Code and rows: `github.com/ravediamond/paper-jepa-world-model` **Reproduce:** `make check`. Every number carries its source file.

1 Introduction

Learned world models summarize a robot's experience in a compact latent space, and a growing body of work proposes to derive useful behaviors and judgments from that space. LeWorldModel (LeWM) [1] trains such a model in two parts: an encoder that maps camera frames to a latent vector, and a predictor that takes the current latent together with a sequence of actions and imagines the latents that follow, both trained end to end from pixels with a joint-embedding predictive (JEPA) loss. Most of the uses proposed for these models rest on the predictor. LeWM itself ranks a policy's candidate action chunks by their imagined outcome; ContactGuard [2] reads the outcome of a grasp before it happens; HWM [3] and H-JEPA [4] plan over several time scales; and other work flags a failure when the observed latent drifts from the predicted one. Each of these papers reports that its use worked on its own robot and its own data.

However, it remains unclear how much of this transfers to the setting in which such a model is most likely to be deployed: a single low-cost arm, 157 teleoperated demonstrations, and a training budget of $3.30. We had one such model, trained on our own arm, and we ran on it every use we could find. Each re-test is a row of a register with a threshold, a dataset, a result and a verdict, and wherever we managed it the threshold and the decision rule were committed to the repository before the result existed. That commit order is what we mean by pre-registration. All 20 rows match a pre-registration record in the commit history; in 19 of them the bar was in place before the result, and the two-level hierarchical progress use had it in the same commit. One further row is not a use of our model but a timing of the pretrained Robo-JEPA [5] and V-JEPA 2.1 [6] encoders on our on-board computer. Of the 20 rows, 14 fail or are no-go (Table 1).

The first version of this paper drew a different conclusion from these failures. We read them as evidence that the latent mostly encodes the arm's pose, in which case a frozen image encoder or the commanded joints should do at least as well. To test this, we ran these controls on 69 policy rollouts, and found instead that the encoder read how the episode went better than either control. We therefore revised the claim, and the present paper is about where the information sits: in the encoder, and not in the predictor.

Throughout the paper we return to one moment, the grasp anchor, 15 frames before the gripper first closes. There the current latent reads the outcome at 0.82 and the imagined latent at 0.70.

For clarity, we note at the outset what is pre-registered and what is not. The controls were added after the data. Each readout is a linear probe with leave-one-session-out (LOSO) cross-validation on the 69 rollouts (44 success, 25 failure, 13 sessions), fit against labels given at rest by a single operator, the author. The model was trained on teleoperation only and never saw these rollouts. No threshold was committed for the control comparisons, so everything this paper says about the split between encoder and predictor is exploratory. Of the 3 register passes, 3 have their bar committed before the result; the register still marks the outcome-readout verdict as retrospective.

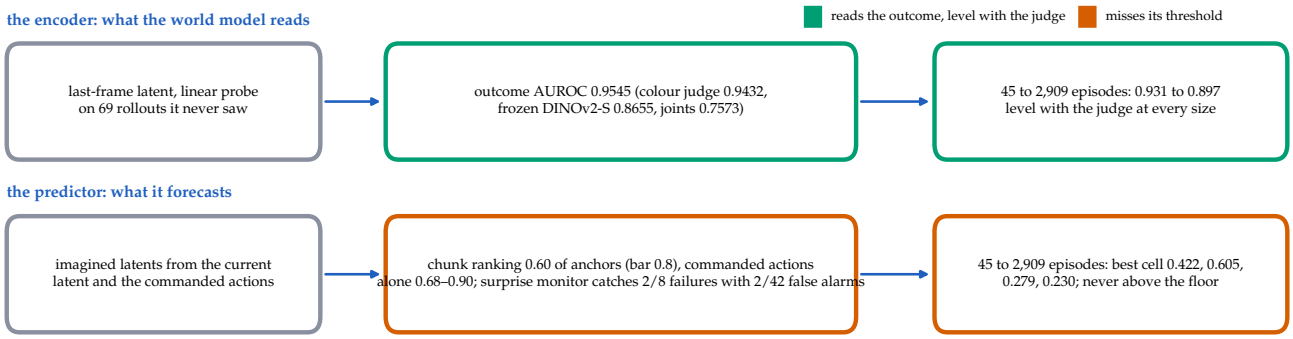


Figure 1: What the world model reads and what it forecasts. The encoder's last-frame latent separates success from failure on rollouts the model never saw about as well as a hand-built colour judge, and it continues to do so when we retrain on 45 to 2,909 episodes. The predictor misses its threshold at the original size and at every other size; the commanded actions rank the executed chunk better than the imagined latents do.

2 Setup

Robot, task and data. The robot is one SO-101 arm [7] driven through LeRobot [8], with a top and a wrist camera. The task is to put a bowl into another bowl. Our model is trained on `ravediamond/so101_bowls_all_trimmed`, split by grid layout into 157 training and 25 validation episodes. The evaluation rollouts come from a separate dataset, `ravediamond/so101-bowls-policy-rollouts`, which consists of SmolVLA [9] runs recorded after our model was trained.

Model and controls. Our model follows the LeWM recipe [1], with a multi-view ViT encoder, an action-conditioned predictor and the SIGReg regulariser. We compare it against four cheaper signals. The frozen DINOv2-S control is a frozen DINOv2 small encoder [10], mean-pooled over the top and wrist views as in DINO-WM [11]; the joints control is the commanded joint vector; the colour judge is a hand-built rule on the final frame with no learning; and policy identity is the checkpoint that ran the episode.

A loading bug and its audit. In October we found that a timing test on the on-board computer had loaded our checkpoint with `strict=False`. The image had transformers 5.5.4, and under that version 192 encoder tensors did not match and were silently dropped, so the test had timed a random encoder. We voided its latents and traced how every other use loaded the model: 26 ran the trained model and 1, the timing test, ran a random one. The world-model block of VLA-JEPA, which one diagnostic below uses, loads all 351 of its keys.

3 The register

Table 1 lists the 20 uses. The takes column records what each one needs from our model: the encoder alone (7), the predictor (8) or both (5).

Among the encoder uses, reading the outcome is the one that works. The outcome readout at rest passes, and at the grasp anchor the current latent reads the grasp at 0.82. Reading progress, in contrast, does not: the four uses that ask the latent for progress or task stage all fail, namely progress monotonicity, two-level hierarchical progress [4], the latent stage tracker and SURVEYOR's latent-distance scale [14]. We also ran the pose probes of LeJEPA [13], and find that a linear readout recovers about half of the joint variance from our encoder (R^2 0.565) and more from the frozen DINOv2-S control (0.867).

The predictor does predict its own latent: at K = 30 its open-loop error is 0.31, against 0.89 for copying the last latent. The uses of this prediction nevertheless fail. In chunk ranking [1] the model puts the executed chunk first in 0.60 of anchors against a threshold of 0.8, whereas the commanded actions alone reach 0.68–0.90; the surprise monitor catches 2/8 at 2/42 failures; and the hierarchical planner [3], the risk head [12], the conformal surprise band, re-ranking and the calibrated reflex all miss their thresholds. Table 1 gives the margins.

4 What the encoder knows

We next ask whether the encoder carries anything that a frozen encoder or the joints do not. Table 2 sets our model beside the four other signals on the same 69 rollouts. Every probe is the same leave-one-session-out logistic regression, fit against the author's labels (`repro/gaps/scripts/analyze_controls.py`):

```
def loso(X, y, groups):
    p = np.zeros(len(y))
    for g in np.unique(groups):
        te = groups == g
        sc = StandardScaler().fit(X[~te])
        m = LogisticRegression(C=1.0, class_weight="balanced",
      max_iter=5000).fit(sc.transform(X[~te]), y[~te])
        p[te] = m.predict_proba(sc.transform(X[te]))[:, 1]
    return p
```

Table 1: The register: every use of our world model re-tested here (with the frozen-DINOv2 and RoboJEPA rows beside it), what it takes from the model, whether its bar came before the result, and the verdict. Read the takes column against the verdict: the predictor uses fail. Bar, result and verdict are quoted from repro/register/expected/FEASIBILITY.md §2.

use	re-tests	takes	bar first	bar	result	verdict	code
open-loop latent prediction	[1]	predictor	yes	beats copy-last; ratios ≥ 1.3	open loop K = 30: 0.31 vs copy-last 0.89; timing shifts 1.99 / 1.58; rank 24–30	**pass**	G0
imagined- vs current-latent outcome probe (first run)	[2]	both	yes	imagined beats current	imagined 0.72, current 0.68; policy identity 0.76	not proven (policy confound)	C1
action-chunk ranking	[1]	predictor	yes	top-1 ≥ 0.8	top-1 best 0.60; action-only 0.68–0.90, regret ≈ 0 (52)	**fail**	G1b
arm-pose linear probe	[1, 11]	encoder	yes	> 0.57	arm-pose R^2 ours 0.57, **DINOv2-S 0.92**	DINOv2-S pass	G1
chunk ranking with a frozen-DINOv2 world model	[11]	predictor	yes	≥ 0.8	DINOv2-S top-1 0.55–0.61	**fail**	G1-W2
progress monotonicity and done detection	this project	encoder	yes	$\rho \leq -0.8$; AUROC ≥ 0.95	monotonicity -0.32 to -0.50; done AUROC ours 0.97 (easy contrast); 32 % false stalls	T1 **fail**, T2 sanity	G6a
outcome readout at rest; imagined-latent anticipation	[2]	both	yes	≥ 0.90; $+0.10$	readout at rest AUROC ours 1.000, DINOv2-S 0.976; anticipation $+0.179$ over policy identity	**pass** (retrospective)	G10
two-level hierarchical progress	[4]	encoder	same commit	≤ -0.80; ≤ 10 %	H-JEPA level 2 ρ -0.727; false stalls 52 %	**fail**	G6b
surprise monitor	this project	predictor	yes	$\geq 5/8$ at $\leq 1/20$	L2 surprise 2/8 at 2/42 (AUROC 0.86); drift 0/8; live reflex 42/42 successes flagged	**fail**	W1-A
latent stage tracker	this project	encoder	yes	≤ 2 %, ≤ 10 %	22.1 % premature, 31.1 % missed	**fail**	W1-B
calibrated surprise reflex	this project	predictor	yes	recall ≥ 50 %	3/25 failures at 1/44	**fail**	G11-B-cal
imagined-latent grasp probe	[2]	both	yes	≥ 0.85, $\geq +0.05$	imagined 0.70 < current 0.82	**fail**	W1-C1
conformal surprise band	this project	predictor	yes	$\leq 1/20$, ≥ 50 %	successes 22.7 % flagged, failures 46 %	**fail**	W1-C2
re-ranking chunk candidates	[1]	predictor	yes	≤ 0.10	regret 0.2631 vs ours 0.2630, action-only -0.0075	**fail**	W1-R
action-conditioned risk head	[12]	both	yes	$\geq 5/8$ at $\leq 1/20$	first version 0.93 withdrawn (time leak); second version 3/8 at 5/42, AUROC 0.79 vs shuffled 0.71	**fail**	REAP
inverse dynamics from a latent pair	this project	encoder	yes	random-init < half ours	latent pair $+0.748$ over one frame, but random-init ours 1.254 vs ours 1.180; beyond state $+0.007$	**not proven**	§30
linear identifiability of the latent	[13]	encoder	yes	$R^2 < 0.80$; ≥ 0.20	ours mean R^2 0.565, anisotropy 0.639; DINOv2-S 0.867 / 0.301	B1–B3 met	LeJEPA
latent-distance scale for plan consumption	[14]	encoder	yes	$\tau \leq 0.40$; $\geq 5/8$ at $\leq 1/20$	τ 1.249 (paper 0.20); monitor 3/8 at 6/42	**fail**	SURVEYOR
hierarchical vs flat prediction horizon	[3]	predictor	yes	crossover in [1, 2] s	no crossover: high level 0.76 vs ours 0.56 at 2 s	H1, H2 **fail**	HWM
on-device latency of RoboJEPA / V-JEPA 2.1	[5, 6]	both	yes	≤ 200 ms; ≥ 15 %	encoder 526 ms (2 views); $\leq$ copy-last + 12 %	**no-go**	RoboJEPA

Table 2: Outcome AUROC of each signal on the same 69 rollouts (exploratory: controls chosen after the data). No judge applies at the anchor.

feature	at rest	grasp anchor (65 scored, 21 fail)
our world model	0.9545 [0.9023, 0.9908]	0.8199 [0.7141, 0.9041]
colour judge (no learning)	0.9432 [0.8919, 0.9878]	—
frozen DINOv2-S	0.8655 [0.7767, 0.9405]	0.7067 [0.5816, 0.8269]
commanded joints	0.7573 [0.6084, 0.8880]	0.5305 [0.3709, 0.6855]
policy identity	0.5882 [0.4084, 0.7515]	—

At rest the interval is a 95 % episode-bootstrap CI. At the anchor it is a bootstrap of the mean over 5 seeds of the out-of-fold AUROC, with episodes resampled jointly across the seeds (2,000 resamples, percentile).

Finding. At rest our model is level with the colour judge (the two CIs overlap) and above both the frozen encoder and the joints. At the grasp anchor, where no judge applies, our model is again above both controls. Notably, the pose probes recover the joints better from the controls than from our encoder, yet the controls read the outcome worse; on these rollouts the encoder therefore carries outcome information that the pose alone does not provide.

The anchor column depends on how the five seeds are combined. Averaging the predictions before scoring gives 0.8377 for our model, 0.7175 for frozen DINOv2-S and 0.5346 for the joints; with seed 0 alone the three signals reach 0.7554, 0.7024 and 0.5574. The gap to the

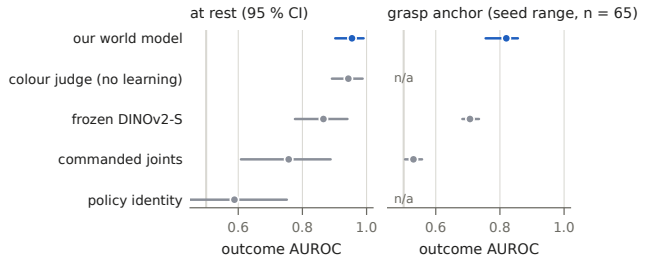


Figure 2: Table 2 as a dot plot. Left: at rest, with the 95 % CI. Right: at the grasp anchor, with the seed range (not a CI).

frozen encoder is smaller with seed 0, but the ordering is unchanged.

On SURVEYOR's scale check the picture reverses and the controls do better than our model: frozen DINOv2-S has τ 0.3601 and the joints τ 0.0265, and both pass, whereas our model has τ 1.2488 and fails (SURVEYOR reports 0.20). Since the task and the rollouts are the same for all three features, we attribute the failure to a property of our latent space rather than of the data.

We note that the register's own outcome-readout row reports 1.000. That number was scored against the colour judge's labels on 87 rollouts; scored against the author's labels on the 69 rollouts, the same readout gives 0.9545.

5 Where the predictor fails

ContactGuard [2] finds that probes on the imagined latent outperform probes on the current latent for grasp tasks. We expected the same at our grasp anchor, and the pre-registered threshold accordingly asked the imagined latent for ≥ 0.85, $\geq +0.05$ over the current one. We find the reverse: with the same encoder and the same anchor, the current latent reads the outcome at 0.82 and the imagined latent at 0.70, so the row fails.

We also ran the distillation probe on our model's inverse-dynamics term, pooled over the 25 teleoperation validation episodes. The trained model scores 0.7476 against 0.7974 for its random initialisation, a difference of -0.0499 with a 95 % paired episode-bootstrap CI of $[-0.7403, +0.6559]$ (`gap5_pooled_rerun.json`). We caution that this term is sensitive to the machine it runs on: the same untrained network scored 1.4383 on one machine and 0.7974 and 0.8042 on two others, which we trace to the BLAS library and to ridge conditioning.

The surprise monitor reaches a failure-class AUROC of 0.775, below its sanity bar. Because a result this low could equally come from a loading or scoring bug, we checked both. The same scorer reproduces the open-loop corruption ratios at K = 30 (`M1_diag_g0_repro.json`); cross-episode gives 3.428 against 3.341 in the original run, and constant-first gives 2.558 against 2.543. We found no bug.

To shed light on the predictor of VLA-JEPA, we ran an action-shuffle check at 5 anchors (Table 3, from `M3_diag_action_shuffle.json`). Action tokens taken from another episode differ from the true ones by 21–28 % in relative L2, yet they move the loss by only about 0.05 %, and zeroing the tokens altogether moves it by about 1.05 % on average. The forward pass is deterministic. In other words, the predictor barely uses its actions. Its failure-class AUROC is 0.501, below the same sanity check, and we leave its table unread under the frozen bar.

Table 3: VLA-JEPA world-model loss at 5 anchors with true, shuffled (Δ shuffled) and zeroed action tokens (exploratory spot check). The token change is the relative L2 of the shuffled tokens.

anchor (episode / frame)	true loss	Δ shuffled (%)	token change (%)	Δ zeroed (%)
8 / 402	1.3031	+0.014	21.1	+2.22
10 / 227	1.4641	+0.088	27.6	+0.50
23 / 327	1.4215	+0.036	21.1	+0.87
59 / 302	1.4637	+0.030	25.5	+1.28
65 / 302	1.5236	−0.070	22.0	−0.38

The outcome-readout gate has one pass that uses the imagined latent: in its anticipation step the imagined latent beat policy identity by +0.179. That bar was set after the result, however, and the pass is exploratory. We therefore reran the idea twice with the threshold committed first, once as the grasp probe and once as the calibrated surprise reflex, and both fail.

6 Does more data change this?

All of the above comes from one model trained on 157 episodes, and the obvious reply to a negative result at this scale is that the model is simply too small. To address this, we trained the same recipe at three other sizes, for the same 81,068 gradient steps as the original model, and ran the same probes on each checkpoint (Table 4). The smallest model sees 45 of the 157 episodes; the original model is the second row and was not retrained; the third adds 653 episodes of a second cubes-and-bowls dataset of ours at the same frame rate; and the fourth adds the 2,099 policy rollouts of ArmnetBench, which come from another lab with other cameras and which we kept at their native 20 Hz, so that the step to the last row changes the data and the setup together. The probes, thresholds and decision rule were committed before any of these runs (`repro/data-scale/PROTOCOL.md`).

At d = 30 the commanded actions rank the executed chunk first at 0.873, and our model is below them at every size by a gap whose CI excludes zero. Its best cell was 0.422 with 45 episodes, 0.605 with 157, then 0.279 and 0.230 after we added the cubes dataset and ArmnetBench; the 0.8 threshold is out of reach at all four sizes. We have not tested why the larger sets did worse. Since the added episodes are other tasks and another rig while the ranking is scored on the same 25 bowls episodes each time, it is possible that the added data takes capacity from the task we score. The surprise monitor caught 1, 2,

Table 4: The probe battery on one model per training-set size. Chunk ranking is the top-1 of 8 candidates at the longest horizon (d = 30) with its gap to the commanded-actions ranker (0.873 top-1 on the same anchors) and the 95 % episode-bootstrap CI of the paired gap. The beyond-state columns come from a ridge probe that predicts the commanded action from the latent now and 5 frames later: the first number is its residual R^2 above the latent now alone or paired with another episode's latent, the second is what the two latents add to the measured joints. The surprise monitor column is the failures caught of 8 at 2 of 42 false interventions. The rest readout is scored on the 69 rollouts against the author's labels.

training episodes (frames)	joints R^2	chunk ranking, d = 30	beyond-state gain / over joints	surprise monitor	rest readout AUROC [CI]
45 (15,032)	0.556	0.403, gap -0.471 [-0.539, -0.405]	1.33 / 0.006	1 of 8	0.931 [0.862, 0.982]
157 (53,015)	0.565	0.605, gap -0.269 [-0.334, -0.204]	0.75 / 0.007	2 of 8	0.955 [0.902, 0.991]
810 (275,015)	0.565	0.273, gap -0.600 [-0.652, -0.545]	0.63 / 0.007	1 of 8	0.956 [0.899, 0.996]
2,909 (1,282,161)	0.573	0.222, gap -0.651 [-0.696, -0.602]	1.17 / 0.009	0 of 8	0.897 [0.813, 0.964]

1 and 0 of 8 failures, where it needed 5.

Neither encoder probe moves with the data. The joints R^2 is 0.556 at the smallest size and 0.573 at the largest, and the rest readout is level with the colour judge (0.943) in every row, since the CI of the paired difference includes zero, although the largest model has the lowest AUROC, 0.897. For the beyond-state probe, the gain of the latent 5 frames later over the latent now alone is at least 0.62, but what it adds to the measured joints is only 0.006 to 0.009 against a threshold of 0.02, and the untrained encoder gains 1.25 from the same extra latent, more than half of any trained one's gain. The verdict is therefore not proven at every size.

We emphasize that this is one run per size with one seed. The decision rule asks for two seeds at the smallest and the largest size, and for a predictor trained on frozen DINOv2-S features at every size as the floor of the surprise monitor; we ran neither. These rows should be read as four runs in which no use of the predictor improves with data, and not as a statement about the model class.

7 Related work

LeWorldModel [1] trains the recipe we use end to end from pixels and plans with it in Push-T and in simulation; we tested its two planning uses on our arm, and both fail. LeJEPA's identifiability analysis [13] predicts that SIGReg latents identify the state up to anisotropy, and our pose probes support this in part. DINO-WM [11] plans on frozen DINOv2 [10] patches, and our frozen control uses the same design. We timed V-JEPA 2.1 [6] and RoboJEPA [5] on our on-board computer, and both are too slow for it. ContactGuard [2] finds imagined-latent probes better than current-latent ones on grasp tasks, whereas on our arm we find the reverse. We also re-tested ReAP's action-conditioned risk [12], the hierarchical planners of HWM [3] and H-JEPA [4], and SURVEYOR's latent-distance scale [14], all of which miss here at the original data scale. Two recent results point in a similar direction: SaiVLA-JEPA [15] reports that a training-only future-latent head adds +0.03 points to control on 10,030 paired episodes, and HIJP [16] finds LeWM below goal-conditioned behaviour cloning in simulation.

8 Limitations

The main limitation of this study is that the probes for the encoder and predictor split were chosen after the data, so the split is exploratory. They are fitted on 69 rollouts that one operator, the author, labelled; the model never saw these rollouts, and every probe is cross-validated by session. Confirming or rejecting the split would require a fresh set of rollouts scored under a threshold committed first, which takes about a day on the arm. The register results come from one model at one data scale, and the data-scale runs add three sizes with one run and one seed each. A second seed at the smallest and the largest size, the frozen-DINOv2-S and random-encoder predictors at each size, the 90-episode size and a successes-only variant of the largest were not run. Three further controls were not run: the surprise monitor and the imagined arm of the grasp probe would each need a predictor trained on frozen DINOv2-S features or on the joints, and progress monotonicity with the controls would need a reimplementation. In the register, the outcome-readout verdict is retrospective and the two-level hierarchical progress use has its bar in the same commit as its result; ordered by commit alone, without the audit, the register reads 13 bar-first and 6 same-commit rows.

9 Conclusion

Conclusion. We launched 20 tests on a latent world model that took $3.30 to train. We found out that the encoder can detect a successful run from a failure after the task is done, without the predictor playing a significant role. Training on 2,909 episodes did not improve this. To rank chunks, the commanded actions, which did not need any model, were better than the predictor. This means that before adding a learned predictor to a pipeline, even on a rig such as ours, its contribution needs to be measured against the encoder by itself and the commanded actions.

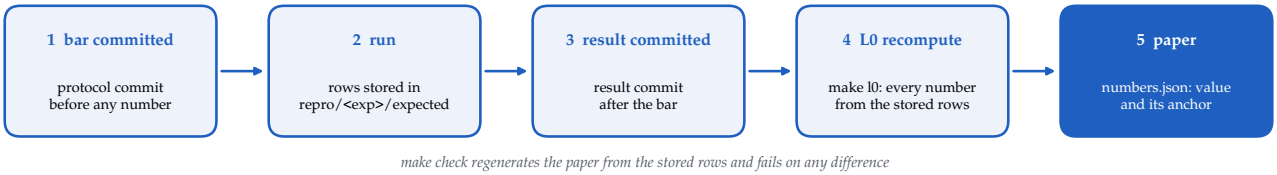


Figure 3: How every number in this paper is made. A protocol with its threshold is committed before any number exists (1). The run stores its rows under repro/ (2). The result is committed after the threshold, and the commit order is audited (3). make l0 recomputes every number from the stored rows (4). make_paper.py writes numbers.json, with each value's anchor, and fills the text from it (5). make check regenerates the paper and fails on any difference.

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